$f: M \to M$ → surjection (injection) → f: iso

↑ Noetherian (Artin~)

Artinian ring

dcc

$\mathbb{Z}$ $(2) \supset (2^2) \supset (2^3) \supset \cdots$

$M/\ker f \cong M$

$\ker T^n \cap \mathrm{Im}\, T^n$

$\ker T^n$

$0 \to L \xrightarrow{f} M \xrightarrow{g} N \to 0$

$0 \to L \xrightarrow{f} M$ ⇔ f: inj

$M \xrightarrow{g} N \to 0$ ⇔ g: sur

M: 노ether ⇔ L, N: 노ether

(Artin) (Artin)

$M/f(L) \cong N$

$M/L \cong N$

Cor M_i: 노ether $(1 \le i \le n)$

$\Rightarrow \bigoplus_{i=1}^{n} M_i$: 노ether

$0 \to M_1 \to \bigoplus_{i=1}^{n} M_i \to \bigoplus_{i=1}^{n-1} M_i \to 0$.